

Mini-Project: Smart Event Booking System (SEBS)

Project Overview:

The *Smart Event Booking System (SEBS)* is a back-end-focused web application that enables users to browse, book, manage, and review various events, including concerts, conferences, workshops, and more. Organisers can create and manage events, while users can register, book tickets, and leave reviews.

This project is designed to demonstrate backend concepts such as RESTful API design, authentication/authorization, data modeling, file uploads, and email notifications using Node.js.

Objectives:

- Apply principles of REST API development using Node.js and Express.
- Implement secure user authentication and authorization using JWT.
- Design scalable and normalized data models using MongoDB.
- Handle media uploads (e.g., event images, ticket PDFs).
- Incorporate email notifications for confirmations and updates.
- Demonstrate separation of concerns via a layered architecture (routes, controllers, services, models).
- Apply best practices for error handling, input validation, and middleware usage.

Key Backend Functionalities:

1. User Module

- **Registration/Login** (JWT-based auth)
- **Profile Management** (view/update personal info)
- **Role-based Access Control** (Admin, Organizer, User)

2. Event Management Module

- **CRUD operations** for events by authorized organisers
- **Event categories & tags**
- **Image Upload** (using `multer` or other)
- **Event status** (upcoming, ongoing, completed)

3. Booking Module

- **Book an event**
- **Cancel booking**
- **Receive confirmation emails** (e.g., via Nodemailer or other)
- **Download tickets (PDF)**

4. Review and Rating Module

- **Submit reviews** only after attending an event
- **Rate events** (1 to 5 stars)
- **Aggregate ratings** for each event

5. Admin Dashboard APIs

- **User management** (ban/unban users, promote to organiser)
- **Event approval system** (events created by organisers need admin approval)
- **View statistics** (number of users, events, bookings, etc.)

6. Notification Module

- **Email Notifications**
 - Upon registration
 - Upon event booking/cancellation
 - Event updates (changes in schedule or venue)

Tech Stack:

- **Node.js + Express.js** (backend framework)
- **MongoDB** (database layer)
- **Mongoose** (ORM/ODM)
- **JWT** (authentication)
- **Nodemailer or other** (email service)
- **Multer or other** (image uploads)
- **PDFKit or other** (ticket generation)
- **Swagger or other** (API documentation)

Expected Deliverables:

- Fully functional backend with REST API
- OpenAPI/Swagger documentation
- Postman collection for testing endpoints
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